

#1 a

```
count=0
while count < 7:
    print("MATHEMATICS")
    count +=1
```

```
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```

b

```
num=2
count=0
while count < 20:
    print(num)
    num += 2
    count += 1
```

```
2
4
6
8
10
12
14
16
18
20
22
24
26
28
30
32
34
36
38
40
```

c

```
num=15
while num<=25:
```

```
print(num)
num+=1
```

```
15
16
17
18
19
20
21
22
23
24
25
```

d

```
n=int(input('enter the number of terms (n):'))
num=5
count=0
while count<n:
    cube=num**3
    print(f'the cube of {num} is {cube}')
    num+=1
    count+=1
```

```
enter the number of terms (n): 10
```

```
the cube of 5 is 125
the cube of 6 is 216
the cube of 7 is 343
the cube of 8 is 512
the cube of 9 is 729
the cube of 10 is 1000
the cube of 11 is 1331
the cube of 12 is 1728
the cube of 13 is 2197
the cube of 14 is 2744
```

e(i)

```
num=105
while num>=7:
    print(num)
    num-=7
```

```
105
98
91
84
```

```
77
70
63
56
49
42
35
28
21
14
7
```

e(ii)

```
n=int(input('enter the number of terms (n):'))
current_term_str=""
count=0
while count<n:
    current_term_str+="2"
    print(current_term_str)
    count+=1
```

enter the number of terms (n): 10

```
2
22
222
2222
22222
222222
2222222
22222222
222222222
2222222222
```

f

```
number_str=input('enter a number:')
num=int(number_str)
sum_of_digits=0
temp_num=num
while temp_num>0:
    digit=temp_num%10
    sum_of_digits+=digit
    temp_num //=10
print(f'the sum of the digits of {num} is {sum_of_digits}')
```

enter a number: 196

the sum of the digits of 196 is 16

#2

a

```
print('first 10 odd numbers')
for i in range(1,20,2):
    print(i)
```

first 10 odd numbers

1
3
5
7
9
11
13
15
17
19

b

```
print('numbers from 3 to 16 and their cubes:')
for num in range(3,17):
    cube=num**3
    print(f'the cube of {num} is {cube}')
```

numbers from 3 to 16 and their cubes:

the cube of 3 is 27
the cube of 4 is 64
the cube of 5 is 125
the cube of 6 is 216
the cube of 7 is 343
the cube of 8 is 512
the cube of 9 is 729
the cube of 10 is 1000
the cube of 11 is 1331
the cube of 12 is 1728
the cube of 13 is 2197
the cube of 14 is 2744
the cube of 15 is 3375
the cube of 16 is 4096

c

```
n=20
print(f'prime numbers up to {n}:')
for num in range(2,n+1):
    is_prime=True
    for i in range(2,int(num**0.5)+1):
```

```

        if num%i ==0:
            is_prime=False
            break
    if is_prime:
        print(num)

```

prime numbers up to 20:

```

2
3
5
7
11
13
17
19

```

d

```

n=7
print(f'multiplication table of {n}:')
for i in range(1,11):
    product=n*i
    print(f'{n}x{i}={product}')

```

multiplication table of 7:

```

7x1=7
7x2=14
7x3=21
7x4=28
7x5=35
7x6=42
7x7=49
7x8=56
7x9=63
7x10=70

```

#3

```

num=74239
n=num
largest=0
smallest=9
while n>0:
    digit=n%10
    if digit > largest:
        largest=digit
    if digit < smallest:
        smallest=digit
    n=n//10

```

```
print('largest digit:',largest)
print('smallest digit:',smallest)
```

```
largest digit: 9
smallest digit: 2
```

#4

```
num=76549
n=num
count=0
reverse=0
while n>0:
    digit=n%10
    reverse=reverse*10+digit
    count+=1
    n=n//10
print('total digits:',count)
print('reversed number:',reverse)
```

```
total digits: 5
reversed number: 94567
```

#5

```
n=int(input('enter number of terms:'))
sum=0
fact=1
print('series terms:')
for i in range(1,n+1):
    fact*=i
    print(fact,end=" ")
    sum+=fact
print('\nsum of the series is:',sum)
```

```
enter number of terms: 10
```

```
series terms:
1 2 6 24 120 720 5040 40320 362880 3628800
sum of the series is: 4037913
```

#6

```
x=float(input('enter the value of x:'))
n=10
result=1.0
term=1.0
for i in range(1,n):
```

```

        term=term*x/i
        result+=term
    print('approximate value of e^x:',result)

enter the value of x: 3

approximate value of e^x: 20.063392857142855

```

#7

```

num=int(input('enter a number:'))
temp=num
sum=0
length=len(str(num))
while temp>0:
    digit=temp%10
    sum+=digit**length
    temp=temp//10
if sum==num:
    print('Armstrong number')
else:
    print('Not an Armstrong number')

```

enter a number: 9474

Armstrong number

#8

```

n=int(input('enter number of rows:'))
for i in range(1,n+1):
    print(" " * (n-i) + "*" * i)

```

enter number of rows: 5

```

    *
   **
  ***
 ****
*****

```

```

n = int(input("Enter number of rows: "))

```

```

for i in range(1, n + 1):
    print(" " * (n - i) + "*" * i)

```

Enter number of rows: 5

```

    *
   * *

```

* * *
* * * *
* * * * *